COLLINEAR FOUR-POINT SUBSETS OF THE GRID $\{0,1,2,3,4\}^n$: A PROOF OF MATHAR'S FORMULAS FOR OEIS A178295

MATH0

ABSTRACT. Let $a(n)$ be the number of 4-element subsets of the grid $\{0,1,2,3,4\}^n \subset \mathbb{R}^n$ that lie on a line. We prove that $a(n) = 9^n/2 + 3 \cdot 7^n/2 - 2 \cdot 5^n$ for every $n \geq 0$, together with the recurrence $a(n) = 21a(n-1) - 143a(n-2) + 315a(n-3)$ for $n \geq 3$ and the generating function $x(-5+41x)/((9x-1)(7x-1)(5x-1))$; these three formulas were conjectured by R. J. Mathar in OEIS A178295 in 2010. The proof writes an ordered collinear 4-tuple as n columns in $\{0,\dots,4\}^4$ that are pairwise affinely related, and a finite classification of the 120 injective columns into 12 classes of size 4 and 36 classes of size 2 gives $24\,a(n) = 12(9^n-5^n)+36(7^n-5^n)$. We found no earlier proof and classify the result as new, although it is elementary and close to the classical count of lines in n-dimensional tic-tac-toe. The theorem is verified in Lean 4 against the frozen, fidelity-reviewed statement that math0 formalized from the OEIS entry, with the finite facts about columns checked by kernel evaluation.

1. INTRODUCTION

Let $G_n = \{0,1,2,3,4\}^n \subset \mathbb{R}^n$. A set $S \subseteq \mathbb{R}^n$ is *collinear* if it is contained in an affine line. Let $\mathcal{Q}_n$ be the set of collinear 4-element subsets of G_n and $a(n) = |\mathcal{Q}_n|$. The OEIS entry A178295 [12], contributed by R. H. Hardin on May 24, 2010 at the suggestion of R. J. Mathar on the Sequence Fans mailing list, lists

$$a(0),\dots,a(9) = 0,\ 5,\ 64,\ 629,\ 5632,\ 48485,\ 410944,\ 3470549,\ 29389312,\ 250334405.$$

The same day Mathar recorded three conjectural formulas: a closed form, a linear recurrence of order 3 and a rational generating function. In the current revision of the entry (#15, July 22, 2025) they are still labelled as a conjecture. We prove all three (Theorem 1). Related entries created the same day carry analogous conjectures of Mathar, among them A178294 [11] (collinear triples in the grid of side 4), A178296 [13] (5-tuples, side 6) and A178297 [14] (6-tuples, side 7).

Prior work. Mathar's manuscript [5], linked from A178294, fits Hardin's tables to Ehrhart polynomials; its abstract calls it a "mostly speculative summary". We could not obtain its full text. Our knowledge of it comes from search snippets of the abstract, seen by the math0 literature referee, and from the following report. Issue #7376 of the GitHub repository `the-omega-institute/trureturing` [9] (opened September 12 and closed September 13, 2026) reports a Lean proof of Mathar's closed form for A178294, collinear triples in $\{0,1,2,3\}^n$. Its literature check found no earlier proof or disproof and describes [5] as presenting the formula as a fit, without proof. It does not mention A178295.

The count of *full* lines is classical: the grid $\{0,\dots,k-1\}^n$, the board of n-dimensional tic-tac-toe of side k, has $((k+2)^n - k^n)/2$ lines containing k grid points. Erdős and Selfridge state this count [1], Golomb and Hales prove it [3, Theorem 1], and Honsberger presents a proof due to L. Moser [4]; for $k=4$, $n=3$ it gives the 76 winning lines of Qubic [8]. For $k=5$ the count is $(7^n - 5^n)/2$, OEIS A081200 [10], where Hardin's comment identifying the sequence with this

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count is also still labelled as a conjecture; math0 has formalized a proof of that identification separately. Mathar's closed form can be written as

$$a(n) = \frac{9^n - 5^n}{2} + 3 \cdot \frac{7^n - 5^n}{2},$$

and Remark 8 explains the two terms geometrically; a reader who knows the word-counting method of [3] can derive the formula quickly in this way. The math0 referees searched arXiv, MathOverflow, Mathematics Stack Exchange, the Sequence Fans mailing list and the tic-tac-toe literature, including [3], and found no published statement of the number of collinear 4-subsets, or of the term $(9^n - 5^n)/2$, and no resolution of the conjecture. We therefore classify Theorem 1 as a *new result*, with two caveats: the argument is short and elementary, and the full text of [5] could not be checked.

Our proof, which is the one formalized, does not use the structure of lattice lines. It writes an ordered collinear 4-tuple as n columns in $\{0, \dots, 4\}^4$, reduces collinearity to a pairwise relation between columns, and classifies the columns by a finite computation (Section 3).

2. Statement

Theorem 1. *Let $a(n)$ be the number of collinear 4-element subsets of $\{0,1,2,3,4\}^n \subset \mathbb{R}^n$. Then:*

(1) $a(n) = 9^n/2 + 3 \cdot 7^n/2 - 2 \cdot 5^n$ *for every $n \geq 0$;*
(2) $a(n) = 21\,a(n-1) - 143\,a(n-2) + 315\,a(n-3)$ *for every $n \geq 3$;*
(3) $\sum_{n \geq 0} a(n)\,x^n = x(-5 + 41x)/\big((9x-1)(7x-1)(5x-1)\big)$ *in the ring $\mathbb{Q}[[x]]$ of formal power series.*

The entry A178295 [12] (offset 0) reads, verbatim (name, author and formula lines):

```
Number of collinear point 4-tuples in a 5 X 5 X 5 X... n-dimensional cubic grid.
_R. H. Hardin_, suggested by R. J. Mathar in the Sequence Fans Mailing List, May 24 2010
Conjecture: a(n) = 9^n/2+3*7^n/2-2*5^n. a(n)= +21*a(n-1) -143*a(n-2) +315*a(n-3). G.f.: x
    *(-5+41*x)/((9*x-1)*(7*x-1)*(5*x-1)). [_R. J. Mathar_, May 24 2010]
```

Lean formalization. math0 formalized the statement in `lean/Math0/Statements/OEIS/A178295.lean`, namespace `Math0.Statements.A178295`. The file was reviewed for fidelity, and its SHA-256 hash `70638d82...bbcc412` recorded, before any proof attempt; the file has not changed since, and this hash is listed in `lean/Math0/Statements/FROZEN.sha256`. The definitions used by the statement, and the statement itself, are (verbatim; docstrings, the two decidability instances, which are obtained by `infer_instance`, and auxiliary declarations are omitted):

```
abbrev Pt (n : ℕ) := Fin n → Fin 5
def coord {n : ℕ} (p : Pt n) (i : Fin n) : ℤ := ((p i : ℕ) : ℤ)
def Coll3 {n : ℕ} (p q r : Pt n) : Prop :=
  ∀ i j : Fin n, (coord q i - coord p i) * (coord r j - coord p j) =
    (coord q j - coord p j) * (coord r i - coord p i)
def IsCollinear {n : ℕ} (s : Finset (Pt n)) : Prop :=
  ∀ p ∈ s, ∀ q ∈ s, ∀ r ∈ s, Coll3 p q r
def a (n : ℕ) : ℕ :=
  ((Finset.univ : Finset (Pt n)).powersetCard 4 |>.filter IsCollinear).card
def statement : Prop :=
  (∀ n : ℕ, (a n : ℚ) = 9 ^ n / 2 + 3 * 7 ^ n / 2 - 2 * 5 ^ n) ∧
  (∀ n : ℕ, 3 ≤ n →
    (a n : ℤ) = 21 * (a (n - 1) : ℤ) - 143 * (a (n - 2) : ℤ) + 315 * (a (n - 3) : ℤ)) ∧
  PowerSeries.mk (fun n => (a n : ℚ)) =
    PowerSeries.X * (-5 + 41 * PowerSeries.X) *
```

```
((9 * PowerSeries.X - 1) * (7 * PowerSeries.X - 1) * (5 * PowerSeries.X - 1))⁻¹
```

Encoding. The grid G_n is `Pt n`; for $n = 0$ it has one point. `Coll3 p q r` says that every 2×2 minor $(q-p)_i(r-p)_j-(q-p)_j(r-p)_i$ vanishes in $\mathbb{Z}$, and `IsCollinear s` requires this for all triples of points of s, repetitions allowed. The statement file proves `isCollinear_iff_collinear`: `IsCollinear s` holds if and only if `Collinear ℝ (toReal '' s)`, where `toReal` views a grid point as a point of $\mathbb{R}^n$ and Mathlib's `Collinear` says that the points lie on one affine line. "Point 4-tuples" are read as unordered sets of four distinct points (`powersetCard 4`); the data force this reading, since $a(1) = 5 = \binom{5}{4}$ and $a(2) = 64 = 25+25+10+4$ counts the four-point subsets of the 5 rows, the 5 columns and the 2 main diagonals, and the 4 diagonals with exactly 4 points. The closed form is stated in $\mathbb{Q}$, so the halves are exact, for all $n \geq 0$ (the offset; the entry gives no range). The recurrence is stated in $\mathbb{Z}$ for $n \geq 3$, where all indices are defined. The generating function uses Mathlib's inverse in $\mathbb{Q}[[x]]$; the denominator has constant coefficient $-1 \neq 0$, so this is the true quotient, and `genFun_iff` restates the clause as $A(x)D(x) = x(-5+41x)$, where $A(x) = \sum a(n)x^n$ and $D(x) = (9x-1)(7x-1)(5x-1)$.

Data check. The definition of `a` is computable but too slow for the kernel. The statement file defines an evaluator `fastA`, which, for each pair of points p, q with p before q in a fixed list, adds $\binom{m}{2}$, where m is the number of points after q on the line pq, and proves $a(n) =$ `fastA`(n) for every n (`a_eq_fastA`). The lemma `a_data` checks $a(0), a(1), a(2) = 0, 5, 64$ by `decide +kernel`. The evaluator does about 5^{3n} work, and a kernel evaluation of $a(3)$ with an evaluator of this kind ran out of memory, so only these three terms are kernel-checked against `a`. The lemmas `closedForm_data` and `recurrence_data` check that the transcribed closed form gives all ten listed terms and that these terms satisfy the transcribed recurrence; they concern the formulas, not `a`. Outside Lean, the math0 referees reproduced $a(1), \dots, a(5) = 5, 64, 629, 5632, 48485$ by brute force.

3. Proof

The proof follows the Lean file (Section 4); each step names its Lean declaration. Write $[5] = \{0,1,2,3,4\}$ and $I = \{0,1,2,3\}$.

3.1. **Columns and compatibility.** A *column* is a map $u\colon I \to [5]$, written $u = (u_0, u_1, u_2, u_3)$ or as a word $u_0u_1u_2u_3$; let $\mathcal{K} = [5]^I$ be the set of the 625 columns and $\mathcal{C} \subseteq \mathcal{K}$ the set of constant columns. A column is *injective* if its four entries are distinct.

Definition 2 (`RR`)**.** Columns u, w are *compatible*, written $u \sim w$, if $(u_b - u_a)(w_c - w_a) = (w_b - w_a)(u_c - u_a)$ for all $a, b, c \in I$; that is, the four points $(u_a, w_a) \in \mathbb{Z}^2$ are collinear.

Lemma 3 (`RR_symm`, `RR_const_right`, `const_of_RR`, `RR_trans`)**.** *Let $u, v, w, t \in \mathcal{K}$.*

(1) *If $u \sim w$ then $w \sim u$.*
(2) *If w is constant, then $u \sim w$.*
(3) *If $w \sim t$, $w_a = w_b$ and $t_a \neq t_b$ for some $a, b \in I$, then w is constant.*
(4) *If $v_{a_0} \neq v_{b_0}$ for some $a_0, b_0 \in I$, $v \sim u$ and $v \sim w$, then $u \sim w$.*

Proof. (1) and (2) are immediate. (3) The instance (a, b, c) of $w \sim t$ reads $0 = (t_b - t_a)(w_c - w_a)$, and $t_b \neq t_a$, so $w_c = w_a$ for every c. (4) Put $D = v_{b_0} - v_{a_0} \neq 0$, $P = u_{b_0} - u_{a_0}$ and $R = w_{b_0} - w_{a_0}$. Subtracting the instances (a_0, b_0, x) and (a_0, b_0, y) of $v \sim u$ gives $D(u_y - u_x) = P(v_y - v_x)$ for all $x, y \in I$, and likewise $D(w_y - w_x) = R(v_y - v_x)$. Hence $D^2(u_b - u_a)(w_c - w_a) = PR\,(v_b - v_a)(v_c - v_a) = D^2(w_b - w_a)(u_c - u_a)$ for all a, b, c, and we cancel $D^2 \neq 0$ in $\mathbb{Z}$. □

3.2. **The finite classification.** Let $r_1, \dots, r_{48}$ be the following injective columns (the list `repL`): $r_1, \dots, r_{12}$ are the 12 permutations of 0123 with first entry 0 or 1, in lexicographic order $(0123, \dots, 1320)$; $r_{13}, \dots, r_{48}$ are the 36 injective columns u whose entries include 0 and 4 and which are lexicographically smaller than $4-u = (4-u_a)_a$, in lexicographic order $(0124, \dots, 2140)$. For $1 \le k \le 48$ let $A_k = \{w \in \mathcal{K} : r_k \sim w\}$.

Lemma 4 (`rep_injB`, `rep_sep`, `cover_check`, `count_check`, `const_check`)**.**

(1) *Each r_k is injective.*
(2) *If $k \neq k'$ then $r_k \not\sim r_{k'}$.*
(3) *Every injective column is compatible with some r_k.*
(4) *$|A_k| = 9$ for $k \le 12$ and $|A_k| = 7$ for $k > 12$.*
(5) *$|\mathcal{C}| = 5$.*

The certificate. In Lean each item is a closed decidable proposition proved by `decide +kernel`: the kernel itself reduces the Boolean test to `true`, with no compiled code and no extra axiom. The test `RRb` evaluates the 64 instances (a, b, c) of the compatibility identity with the statement file's `minorB`, which compares two sums of products in $\mathbb{N}$ (correct by `minorB_iff`), and `RRb_iff` proves that `RRb` decides $\sim$. Items (3)–(5) run over the statement file's list `L 4` of all 625 columns, which `mem_L` and `nodup_L` show to be complete and duplicate-free; `card_filter_col` turns counts along `L 4` into cardinalities of finite sets. The kernel thus tests at most $48 \cdot 625$ pairs of columns per item.

Proof by hand. Lean uses only the computation. Let u be injective. If $w = \alpha u + \beta$ with $\alpha, \beta \in \mathbb{Q}$, both sides of the identity defining $u \sim w$ equal $\alpha(u_b - u_a)(u_c - u_a)$, so $u \sim w$. Conversely, if $u \sim w$, the instances (a_0, b_0, c) with $a_0 \neq b_0$ fixed give $w = \alpha u + \beta$. So $A_u = \{w : u \sim w\}$ consists of the 5 constant columns ($\alpha = 0$) and of injective columns $\alpha u + \beta$ whose span $\max - \min$ is $|\alpha|$ times that of u; four distinct values in [5] have span 3 or 4. If u has span 3, its values are $m, \dots, m+3$ with $m \in \{0, 1\}$, and $|\alpha| \in \{1, 4/3\}$; $|\alpha| = 4/3$ is impossible, since the entries of w where u takes the values m and $m+1$ would differ by $4/3$. So $w \in \{u - m,\ u - m + 1,\ m + 3 - u,\ m + 4 - u\}$. If u has span 4, its values are $0 < x < y < 4$ and $0, 4$, and $|\alpha| \in \{1, 3/4\}$; $|\alpha| = 3/4$ is impossible, since the entries where u takes the values 0 and x would differ by $3x/4 \notin \mathbb{Z}$. So $w \in \{u, 4 - u\}$. By Lemma 3(1),(4), $\sim$ is an equivalence relation on the 120 injective columns, with classes of size 4 (the $2 \cdot 4! = 48$ columns of span 3) and 2 (the $3 \cdot 4! = 72$ columns of span 4), so 12 and 36 classes. The r_k are exactly the lexicographically smallest members of the 48 classes (in a class of span 3 the smallest member takes the values $0, 1, 2, 3$, and of v and $3 - v$ it is the one with first entry at most 1; in a class $\{u, 4 - u\}$ it is the smaller of the two). This gives (1)–(3); (4) follows from $|A_k| = 5 + 4$ or $5 + 2$, and (5) is clear. □

3.3. **From sets to column families.** Let $E(n)$ be the set of injective maps $f \colon I \to G_n$ such that $\texttt{Coll3}(f(a), f(b), f(c))$ for all $a, b, c \in I$ (`E`). Let $V(n)$ be the set of families $g = (g_i)_{i<n}$ of columns such that $g_i \sim g_j$ for all i, j and the rows are distinct: for all $a \neq b$ in I some i has $g_i(a) \neq g_i(b)$ (`V`).

Lemma 5 (`fiber_card`, `card_E`, `card_E_V`)**.** $|V(n)| = |E(n)| = 24\, a(n)$.

Proof. For $f \in E(n)$ the image $f(I)$ has 4 elements and is collinear, since any three of its points are $f(a), f(b), f(c)$. Conversely, if $s \in \mathcal{Q}_n$, every injective f with $f(I) = s$ lies in $E(n)$. So $f \mapsto f(I)$ maps $E(n)$ onto $\mathcal{Q}_n$, and each fibre is the set of bijections $I \to s$, in bijection with the embeddings $I \hookrightarrow s$, of which there are $4! = 24$. Hence $|E(n)| = 24\, a(n)$. Transposition $f \mapsto g$, $g_i(a) = f(a)_i$, is a bijection from maps $I \to G_n$ to families of n columns. The instance (i, j) of $\texttt{Coll3}(f(a), f(b), f(c))$ is literally the instance (a, b, c) of $g_i \sim g_j$, and f is injective exactly when the rows of g are distinct. Hence $|E(n)| = |V(n)|$. □

3.4. **Decomposition.** For $1 \le k \le 48$ let $W(n,k) = A_k^n \setminus \mathcal{C}^n$, the families of n columns, each compatible with r_k, not all constant (`W`).

Proposition 6 (`V_eq`, `W_disj`)**.** *$V(n)$ is the disjoint union of $W(n,1),\dots,W(n,48)$.*

Proof. $V(n) \subseteq \bigcup_k W(n,k)$. Let $g \in V(n)$. Rows 0 and 1 are distinct, so some column g_{i_0} is not constant. It is injective: if $g_{i_0}(a) = g_{i_0}(b)$ with $a \ne b$, some g_j has $g_j(a) \ne g_j(b)$, and Lemma 3(3) applied to $g_{i_0} \sim g_j$ makes g_{i_0} constant. By Lemma 4(3), $r_k \sim g_{i_0}$ for some k. For every j, Lemma 3(4) through the non-constant g_{i_0}, applied to $g_{i_0} \sim r_k$ and $g_{i_0} \sim g_j$, gives $r_k \sim g_j$. So $g \in W(n,k)$.

$W(n,k) \subseteq V(n)$. Let $g \in W(n,k)$. Since r_k is injective, Lemma 3(4) through r_k gives $g_i \sim g_j$ for all i,j. Choose a non-constant g_{i_0}. If $a \ne b$ and $g_i(a) = g_i(b)$ for all i, Lemma 3(3) applied to $g_{i_0} \sim r_k$, with $r_k(a) \ne r_k(b)$, makes g_{i_0} constant, a contradiction.

Disjointness. If $g \in W(n,k) \cap W(n,k')$, choose a non-constant g_{i_0}; Lemma 3(4) through g_{i_0} gives $r_k \sim r_{k'}$, so $k = k'$ by Lemma 4(2). □

Lemma 7 (`card_W`, `card_V`)**.** *$|W(n,k)| = |A_k|^n - 5^n$, and $|V(n)| = 12(9^n - 5^n) + 36(7^n - 5^n)$.*

Proof. By Lemma 3(2), $\mathcal{C}^n \subseteq A_k^n$, so $|W(n,k)| = |A_k|^n - |\mathcal{C}|^n$. Use Lemma 4(4),(5) and sum over k by Proposition 6. □

For $n = 0$ all these sets are empty, in agreement with $a(0) = 0$; no separate case is needed.

3.5. **Proof of Theorem 1.** *(1)* (`a_mul`, `closed`). Lemmas 5 and 7 give $24\,a(n) = 12(9^n - 5^n) + 36(7^n - 5^n)$. In Lean this holds in $\mathbb{N}$ with truncated subtraction; since $5^n \le 7^n \le 9^n$ it casts to $\mathbb{Q}$, and dividing by 24 gives $a(n) = (9^n - 5^n)/2 + 3(7^n - 5^n)/2 = 9^n/2 + 3 \cdot 7^n/2 - 2 \cdot 5^n$.

(2) (`result`). Since $x^3 - 21x^2 + 143x - 315 = (x-5)(x-7)(x-9)$, each of $5^n, 7^n, 9^n$, and hence $a(n)$ by (1), satisfies $s(n) = 21s(n-1) - 143s(n-2) + 315s(n-3)$. In Lean one writes $n = m + 3$, casts to $\mathbb{Q}$, substitutes (1) and closes the goal by `ring`.

(3) (`genFun_iff`, `result`). $D(x) = -1 + 21x - 143x^2 + 315x^3$ is invertible in $\mathbb{Q}[[x]]$, and the claim is equivalent to $A(x)D(x) = -5x + 41x^2$. The coefficient of x^n in $A(x)D(x)$ is $-a(n) + 21a(n-1) - 143a(n-2) + 315a(n-3)$, terms with negative index omitted. For $n = 0, 1, 2$ it is 0, $-5 + 0 = -5$ and $-64 + 105 - 0 = 41$, using $a(0), a(1), a(2) = 0, 5, 64$ from (1); for $n \ge 3$ it is 0 by (2) (in Lean, again by substituting (1) and `ring`). □

Remark 8 (geometric meaning; not formalized)**.** If $g \in W(n,k)$ with r_k of span 3, the four points are $x(0), x(1), x(2), x(3)$ for a map $t \mapsto x(t)$ each of whose coordinates is constant, equal to t or $t+1$, or equal to $3 - t$ or $4 - t$: 9 choices per coordinate, not all constant, so the four points are consecutive lattice points on a line. Counting such words up to reversal, as Golomb and Hales count lines [3], gives $(9^n - 5^n)/2$ sets of four consecutive collinear points. If r_k has span 4, the points are four of the five points of a full line, including both endpoints; there are $(7^n - 5^n)/2$ full lines [1, 3], each with three such subsets. This matches $12(9^n - 5^n)/24$ and $36(7^n - 5^n)/24$. A math0 referee derived the formula independently in this way, using that a line whose primitive direction has a coordinate of absolute value at least 2 meets G_n in at most 3 points. The same count for collinear k-subsets of $\{0,\dots,k\}^n$ suggests $\big((k+5)^n - (k+1)^n + (k-1)((k+3)^n - (k+1)^n)\big)/2$, which agrees with Mathar's conjectured closed forms in A178294, A178296 and A178297 ($k = 3, 5, 6$); we have not formalized these cases.

4. Machine verification

The proof is in `lean/Math0/Frontier/OEIS/A178295.lean` of the math0 repository (`https://github.com/Foundation-0/math0`). It imports the frozen statement file (Section 2) and

the audit commands of `lean/Math0/Audit.lean`. The main declaration is `theorem result : statement` in the namespace `Math0.Frontier.A178295`; the auxiliary declarations of this file named in Section 3 (all except those attributed there to the statement file) are `private`, and the file ends with

```
#math0_settles Math0.Frontier.A178295.result Math0.Statements.A178295.statement
```

Neither file contains `sorry`, `admit`, `axiom`, `native_decide`, `implemented_by`, `extern`, `unsafe`, `opaque` or `set_option`; the finite checks of Lemma 4 run within the default limits.

The command `#math0_settles`, used for statements that math0 formalizes itself, fails unless (i) `statement` is a `Prop`-valued definition without universe parameters; (ii) `statement` and `result` depend on no axioms besides `propext`, `Classical.choice` and `Quot.sound`, which excludes `sorryAx`, the axiom `Lean.ofReduceBool` behind `native_decide`, and custom axioms; and (iii) the type of `result` is syntactically `statement` or its negation, up to metadata, with no unfolding. Compiling the file with `lake env lean` prints the following, and no warnings:

```
math0 audit: 'Math0.Statements.A178295.statement' depends on axioms [Classical.choice, Quot.sound, propext]
math0 audit: 'Math0.Frontier.A178295.result' depends on axioms [Classical.choice, Quot.sound, propext]
math0 statement check: 'Math0.Frontier.A178295.result' proves 'Math0.Statements.A178295.statement'
```

Toolchain and reproduction. Lean 4 [7] `v4.33.1` and Mathlib [6] `v4.33.1`, pinned in `lean/`. The repository also pins formal-conjectures [2] at commit `c77fdd4`; this result does not use it, since the statement is math0's own formalization of an OEIS conjecture. To reproduce, run `scripts/verify.sh` from the repository root; it fetches the Mathlib cache, runs `lake build` on all math0 results (re-running every audit command) and checks that the pinned dependencies are unmodified. The single file is checked by `lake env lean Math0/Frontier/OEIS/A178295.lean` in `lean/`; this takes about two minutes, nearly all of it in the `decide +kernel` checks.

References

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